JEE Mathematics: Quick Review with Grading and Practice Map - Differential Equation (Batch XII)

Grade shows how often the item is tested in the JM exercise of this batch, the JA exercise and the JEE Main Question Bank taken together (144 questions; a question that appears in two of these is counted once): High = 13 or more questions, Medium = 1 to 12 questions, Low = none. Low only means the item is not tested in these questions; it can still be important for understanding. Practice map lists the questions that test the item in the sheet exercises (J-I, J-II, J-M, J-A, C-I, C-II), the DPPs and the Question Bank (QB). 'Main concept' = the question is mainly about this item; 'Also used in' = the question needs it as a secondary step.

Introduction

Question 01

What do you mean by ordinary differential equation and partial differential equation?

Model Answer

An equation containing the independent variable, the dependent variable and derivatives of the dependent variable is a differential equation (DE).

Ordinary: the dependent variable depends on ONE independent variable, e.g. $\frac{dy}{dx} + xy = \sin x$.

Partial: on two or more, with partial derivatives, e.g. $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0$.

JEE uses only ordinary DEs.

Grade - **Low**

Practice map, main concept: none

Also used in: none

Question 02

What do you mean by general solution, complete integral, complete primitive?

Model Answer

A relation between x and y (free of derivatives) that satisfies the DE is a solution (primitive). The general solution contains as many independent arbitrary constants as the order of the DE.

General solution = complete integral = complete primitive.

Ex: $\frac{dy}{dx} = 2x \Rightarrow y = x^2 + c$ (general solution).

Grade - **Low**

Practice map, main concept: Sheet: J-I 5

Also used in: none

Question 03

Define particular solution of a differential equation.

Model Answer

A solution obtained from the general solution by giving particular values to the arbitrary constants (usually from an initial condition $y(x_0) = y_0$).

Ex: $y = x^2 + c$ through $(2, 5)$: $c = 1$, so $y = x^2 + 1$.

Grade - **Low**

Practice map, main concept: none

Also used in: Sheet: J-I 48

Question 04

How to fix the constant when, instead of $y(x_0)$, a limit or boundedness condition is given (e.g. y bounded, $y \to 0$ as $x \to \infty$)?

Model Answer

Solve generally, then take the limit of the general solution: a term like Ce^{kx} ($k > 0$, $x \to \infty$) must have $C = 0$; a condition like $\lim_{x\to 0} xy = 1$ gives C directly.

Ex: $y' - y = -e^{-x}$: $y = \frac{e^{-x}}{2} + Ce^{x}$; $y \to 0$ as $x \to \infty \Rightarrow C = 0$.

Grade - **Medium**

Practice map, main concept: none

Also used in: Sheet: J-II 49; J-M 14, 63; C-I 4; C-II 6 | Question Bank: 125, 140

Order and Degree of a Differential Equation

Question 05

Define order of a differential equation and how to find it?

Model Answer

Order = order of the highest order derivative present in the DE.

Ex: $\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^3 = y^2 + \sin x$: order 2.

Grade - **Low**

Practice map, main concept: none

Also used in: Sheet: J-I 1, 4, 6, 10

Question 06

Define degree of a differential equation and how to find it?

Model Answer

First make the DE a polynomial in all the derivatives (remove radicals and fractional powers of derivatives). Degree = power of the highest order derivative.

Ex: $\frac{d^2y}{dx^2} = \left(1 + \left(\frac{dy}{dx}\right)^2\right)^{1/3}$: cube both sides, order 2, degree 3.

Grade - **Low**

Practice map, main concept: Sheet: J-I 1

Also used in: Sheet: J-I 6, 9-10; J-II 51

Question 07

Does every differential equation have a degree and order?

Model Answer

Order always exists. Degree is NOT defined when a derivative occurs inside a transcendental function (sin, $e^{(\ )}$, ln, ...) and the DE cannot be made a polynomial in derivatives.

Ex: $\frac{d^2y}{dx^2} + \sin\left(\frac{dy}{dx}\right) = x$: order 2, degree not defined. But $\sin^{-1}\left(\frac{dy}{dx}\right) = x + 2 \Rightarrow \frac{dy}{dx} = \sin(x + 2)$: degree 1.

Grade - **Low**

Practice map, main concept: none

Also used in: none

Formation of Differential Equation

Question 08

How to check whether arbitrary constants in the equation of a family of curves are independent?

Model Answer

Constants are independent when the equation cannot be rewritten with fewer constants. Combine constants first: $c_1 e^{x+c_2} = (c_1 e^{c_2})e^x = Ae^x$.

Ex: $ax + by + c = 0$: 2; $y^2 = 4A(x + A)$: 1; $y^2 = 4A(x + B)$: 2; $y = c_1 e^{x+c_2}$: 1.

Grade - Low

Practice map, main concept: Sheet: J-I 4

Also used in: Sheet: J-I 13; C-II 23

Question 09

How to form a differential equation if the equation of a family of curves is given?

Model Answer

Count the independent constants n; differentiate n times; eliminate the constants. The DE has order n.

Ex: $y = Ae^{2x} + Be^{-2x} + C$: $\frac{d^3y}{dx^3} = 4\frac{dy}{dx}$.

Grade - Medium

Practice map, main concept: Sheet: J-I 6-10, 13; J-II 1; C-II 23 | Question Bank: 112

Also used in: Question Bank: 107

Question 10

How to form a differential equation if information about the family of curves is provided instead of its equation?

Model Answer

Write the general equation of the family from the description (with the fewest constants), then differentiate and eliminate as usual.

Ex: circles touching the x-axis at O: $x^2 + y^2 - 2ay = 0 \Rightarrow (x^2 - y^2)\frac{dy}{dx} = 2xy$.

Grade - Medium

Practice map, main concept: Sheet: J-I 2-3, 11-12; J-M 19, 57; J-A 5 | Question Bank: 107

Also used in: none

Elementary Types

Question 11

What are the various types of differential equations?

Model Answer

First order, first degree types: variable separable (incl. $f(ax + by + c)$, polar substitution), homogeneous, reducible to homogeneous, linear (in y or in x), Bernoulli / reducible to linear, exact differential.

Check in this order: separable? homogeneous? linear in y or x? Bernoulli / substitution? exact grouping?

Grade - Low

Practice map, main concept: Sheet: C-II 31

Also used in: none

Question 12

How to solve simple second order differential equations?

Model Answer

$y'' = f(x)$: integrate twice (two constants). y absent: put $p = \frac{dy}{dx}$. $y'' = f(y)$: multiply by $2y'$: $(y')^2 = 2\int f(y)\,dy + C$.

Ex: $y'' = y \Rightarrow (y')^2 = y^2 + C$; $y'' = 6x \Rightarrow y = x^3 + c_1x + c_2$.

Grade - Medium

Practice map, main concept: Sheet: J-I 17 | Question Bank: 127

Also used in: none

Variable Separable

Question 13

How to solve variable separable differential equations?

Model Answer

Write it as $f(x)\,dx = g(y)\,dy$ and integrate: $\int f(x)\,dx = \int g(y)\,dy + c$.

Ex: $x(y^2+1)dx + y(x^2+1)dy = 0 \Rightarrow (x^2+1)(y^2+1) = C$.

Grade - High

Practice map, main concept: Sheet: J-I 14-16, 20-22, 28-29, 31-32, 37; J-II 2-4, 6-7, 14; J-M 1, 6, 8, 11-13, 21, 23, 35, 42, 44, 49; J-A 8, 12-13, 15, 17; C-I 6, 18, 21; C-II 2, 22, 24, 32-33 | Question Bank: 98, 105, 111, 114, 117-118, 121, 125, 138, 140, 143

Also used in: Sheet: J-I 17, 24-27, 33-36; J-II 5, 8-13, 15, 34, 39, 52; J-M 20, 27-28; J-A 14; C-I 2, 5, 7, 11-12; C-II 4, 7, 12, 14, 16, 20, 31 | Question Bank: 102, 116, 124, 127, 130, 137

Question 14

How to solve differential equation questions by forming the differential equation with the help of given information about the curve by assuming a point (x, y) on the curve?

Model Answer

Take $P(x, y)$, $m = \frac{dy}{dx}$. Tangent $Y - y = m(X - x)$: x-intercept $x - \frac{y}{m}$, y-intercept $y - mx$. Normal $Y - y = -\frac{1}{m}(X - x)$: x-intercept $x + my$.

Subtangent $\left|\frac{y}{m}\right|$, subnormal $|ym|$, tangent length $\left|\frac{y}{m}\right|\sqrt{1+m^2}$, normal length $|y|\sqrt{1+m^2}$. Translate the condition into a DE, solve, use the given point.

Ex: tangent segment between the axes bisected at P: $\frac{dy}{dx} = -\frac{y}{x} \Rightarrow xy = c$; through $(2, 3)$: $xy = 6$.

Grade - Medium

Practice map, main concept: Sheet: J-I 24-25, 27, 39, 41; J-II 10, 12-13, 17-20, 22, 25, 29-31, 39-40, 50-52, 56; J-M 27; J-A 14; C-I 14; C-II 7, 10, 12, 14-15, 18 | Question Bank: 137, 145

Also used in: none

Question 15

How to solve differential equation questions by forming the differential equation if an identity in a definite integral is given?

Model Answer

Differentiate both sides (Leibniz rule) to get a DE; the initial condition comes from putting the lower limit in the identity.

Ex: $f(x) = \int_0^x e^t f(t)\,dt + e^x \Rightarrow f' = e^x(f+1),\ f(0) = 1 \Rightarrow f(x) = 2e^{e^x - 1} - 1$.

Grade - **Medium**

Practice map, main concept: Sheet: J-II 15, 34; J-M 20, 25, 55, 62; C-I 5, 7; C-II 11, 28-30 | Question Bank: 89, 92, 102, 116, 130

Also used in: Sheet: J-I 25 | Question Bank: 90

Question 16

How to solve rate problems (growth and decay, Newton's law of cooling, mixture in a tank) by forming a differential equation?

Model Answer

Write "rate of change = given law": growth/decay $\frac{dN}{dt} = kN \Rightarrow N = N_0 e^{kt}$; cooling $\frac{dT}{dt} = -k(T - T_s)$; mixture $\frac{dQ}{dt} = (\text{rate in}) - \frac{Q}{V(t)} \cdot (\text{outflow})$. Two data fix C and k.

Ex: $N(0) = 1000$, 20% rise in 2 h: $N = 1000\left(\frac{6}{5}\right)^{t/2}$; doubling time $t = \frac{2\ln 2}{\ln(6/5)}$.

Grade - **Low**

Practice map, main concept: Sheet: J-I 26, 33-36; J-II 8-9, 36-37; C-I 2; C-II 16-17, 20

Also used in: none

Question 17

When to assume the integration constant as $\ln c$?

Model Answer

When the other terms after integration are logarithms: write $+\ln c$ so that the answer can be combined into one log and written without logs.

Ex: $\frac{dy}{y} = \frac{dx}{x} \Rightarrow \ln|y| = \ln|x| + \ln c \Rightarrow y = cx$.

Grade - **Medium**

Practice map, main concept: none

Also used in: Sheet: J-M 27

Question 18

What do you mean by initial ordinate of a line (tangent)?

Model Answer

Initial ordinate = the y-intercept of the line. For the tangent at $P(x, y)$: $y - x\frac{dy}{dx}$.

Grade - **Low**

Practice map, main concept: none

Also used in: Sheet: J-II 31, 40

Question 19

What do you mean by polar radius of a point?

Model Answer

Polar radius (radius vector) of $P(x, y)$ = distance $OP = \sqrt{x^2 + y^2}$; the polar angle is θ with $\tan\theta = \frac{y}{x}$.

Ex: normal length $= OP$: $y^2(1 + m^2) = x^2 + y^2 \Rightarrow m = \pm\frac{x}{y} \Rightarrow x^2 \mp y^2 = C$ (rectangular hyperbola or circle).

Grade - Low

Practice map, main concept: none

Also used in: Sheet: J-I 39; J-II 12, 56

Question 20

How to solve $\frac{dy}{dx} = f(ax + by + c)$?

Model Answer

Put $t = ax + by + c$: $\frac{dt}{dx} = a + bf(t) \Rightarrow \int \frac{dt}{a+bf(t)} = x + C$.

Ex: $\frac{dy}{dx} = (x + y)^2$: $t = x + y$, $\frac{dt}{dx} = 1 + t^2 \Rightarrow \tan^{-1}(x + y) = x + C$.

Grade - Medium

Practice map, main concept: Sheet: J-I 18-19; J-II 5, 24; C-I 12 | Question Bank: 124, 134, 141

Also used in: none

Question 21

How to solve $\frac{dy}{dx} = \frac{a_1x+b_1y+c_1}{a_2x+b_2y+c_2}$ if $a_2 + b_1 = 0$?

Model Answer

Cross-multiply; the terms $a_2x\,dy + (-b_1)y\,dx$ combine to $a_2\,d(xy)$; integrate term by term.

Ex: $\frac{dy}{dx} = \frac{4x-3y}{3x-2y}$: $3\,d(xy) - 2y\,dy - 4x\,dx = 0 \Rightarrow 3xy - y^2 - 2x^2 = c$.

Grade - Medium

Practice map, main concept: Sheet: J-M 34

Also used in: none

Question 22

What are the two polar substitutions and derive their corresponding results involving differentials?

Model Answer

$x = r\cos\theta$, $y = r\sin\theta$: $x\,dx + y\,dy = r\,dr$, $x\,dy - y\,dx = r^2 d\theta$.

$x = r\sec\theta$, $y = r\tan\theta$: $x\,dx - y\,dy = r\,dr$, $x\,dy - y\,dx = r^2\sec\theta\,d\theta$.

(From $x^2 + y^2 = r^2$ and $\tan\theta = \frac{y}{x}$; resp. $x^2 - y^2 = r^2$ and $\sin\theta = \frac{y}{x}$.)

Grade - Low

Practice map, main concept: Sheet: J-II 11

Also used in: Sheet: J-II 56

Question 23

When to use polar substitutions in differential equation questions?

Model Answer

When the DE contains $x\,dx \pm y\,dy$, $x\,dy - y\,dx$ together with $x^2 \pm y^2$ or $\frac{y}{x}$.

Ex: $\frac{x\,dx+y\,dy}{\sqrt{x^2+y^2}} = \frac{y\,dx-x\,dy}{x^2}$: $dr = -\sec^2\theta\,d\theta \Rightarrow \sqrt{x^2 + y^2} + \frac{y}{x} = C$.

Grade - Low

Practice map, main concept: none

Also used in: Sheet: J-II 11

Homogeneous Differential Equation

Question 24

How to check whether a given differential equation is a homogeneous differential equation?

Model Answer

$\frac{dy}{dx} = \frac{f(x,y)}{g(x,y)}$ is homogeneous if f, g are homogeneous of the same degree: $f(\lambda x, \lambda y) = \lambda^n f(x,y)$; equivalently $\frac{dy}{dx} = F\left(\frac{y}{x}\right)$.

Ex: $(x-y)\frac{dy}{dx} = x + 2y$ is homogeneous (both sides degree 1).

Grade - Low

Practice map, main concept: Sheet: J-II 48

Also used in: none

Question 25

What substitution is to be done in homogeneous differential equations?

Model Answer

Put $y = vx$: $\frac{dy}{dx} = v + x\frac{dv}{dx}$, so $\int \frac{dv}{F(v)-v} = \ln|x| + c$. (If $\frac{dx}{dy} = G\left(\frac{x}{y}\right)$ is simpler, put $x = vy$.)

Ex: $\frac{dy}{dx} = \frac{x+y}{x}$: $x\frac{dv}{dx} = 1 \Rightarrow y = x\ln|x| + cx$.

Grade - Medium

Practice map, main concept: Sheet: J-I 23, 30, 38, 40, 59; J-II 16, 21; J-M 22, 41, 48, 61; J-A 2, 6; C-I 9 | Question Bank: 115, 120, 123, 144

Also used in: Sheet: J-I 39, 41; J-II 17-20, 22, 25; J-M 9, 59; C-I 14; C-II 15

Equations reducible to Homogeneous Differential Equation

Question 26

How to solve the equation $\frac{dy}{dx} = \frac{L_1}{L_2}$ if L_1 and L_2 are intersecting lines?

Model Answer

$L_i = a_i x + b_i y + c_i$. Find the point of intersection (h,k); put $x = X + h$, $y = Y + k$: $\frac{dY}{dX} = \frac{a_1X + b_1Y}{a_2X + b_2Y}$ (homogeneous), then $Y = vX$.

If $L_1 \parallel L_2$ put $t = a_1x + b_1y$ (#17); if $a_2 + b_1 = 0$ group into $d(xy)$ (#18).

Ex: $\frac{dy}{dx} = \frac{x+2y-4}{2x+y-5}$: lines meet at $(2,1)$; $X = x-2$, $Y = y-1$: $\frac{dY}{dX} = \frac{X+2Y}{2X+Y}$.

Grade - Medium

Practice map, main concept: Sheet: J-II 23; J-M 9, 59 | Question Bank: 142

Also used in: Sheet: J-A 6

Linear Differential Equation

Question 27

How to check whether a given differential equation is a linear differential equation?

Model Answer

Linear: the dependent variable and all its derivatives occur only in the first degree and are never multiplied together (no y^2, yy', $\sin y$, ...).

Ex: $\frac{dy}{dx} + y\tan x = \sec x$ is linear; $\frac{dy}{dx} + y^2 = x$ is not.

Grade - Low

Practice map, main concept: Sheet: J-I 42

Also used in: none

Question 28
What is the general form of a linear differential equation?

Model Answer

$$a_0(x)\frac{d^n y}{dx^n} + a_1(x)\frac{d^{n-1}y}{dx^{n-1}} + \cdots + a_n(x)\, y = \phi(x)$$

The coefficients $a_0, \ldots, a_n$ and ϕ are functions of x only.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 29
Are all differential equations of degree one linear?

Model Answer
No. A linear DE always has degree 1, but a degree-1 DE need not be linear.
Ex: $\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^3 + y^2 = 0$: degree 1, not linear.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 30
What is the standard form of a first order linear differential equation?

Model Answer

$$\frac{dy}{dx} + P(x)\, y = Q(x)$$

Linear in x: $\frac{dx}{dy} + P(y)\, x = Q(y)$ (use it when y appears non-linearly but x linearly).

Grade - Medium
Practice map, main concept: none
Also used in: Sheet: J-I 44, 47, 51, 57; J-II 30, 33; J-M 18, 26, 36; C-II 5, 13, 31 | Question Bank: 97, 99, 148

Question 31
How to solve a first order linear differential equation?

Model Answer

$$\text{IF} = e^{\int P\, dx}, \qquad y \cdot \text{IF} = \int Q \cdot \text{IF}\, dx + C$$

Bring to standard form, multiply by IF; the left side becomes $\frac{d}{dx}(y \cdot \text{IF})$; integrate. Use $e^{\ln g} = g$.
Ex: $\frac{dy}{dx} + \frac{y}{x} = x^2$: IF $= x$, $xy = \frac{x^4}{4} + C$.

Grade - High
Practice map, main concept: Sheet: J-I 43-53, 58; J-II 26-28, 32-33, 35, 38, 49, 53; J-M 2-3, 16-18, 24, 26, 32-33, 36, 38, 40, 43, 45-47, 50-52, 54, 56, 58, 60, 63; J-A 1, 3-4, 7, 16, 18-20; C-I 1, 4, 8, 15-16, 19-20; C-II 6, 19, 25-27 | Question Bank: 90-91, 94-97, 99, 101, 103-104, 106, 108-109, 119, 122, 126, 128-129, 131-133, 135-136, 147-148
Also used in: Sheet: J-I 36, 54-55, 57; J-II 29-31, 36, 40-42, 45-47, 50; J-M 4, 10, 15, 25; J-A 9; C-I 13, 17; C-II 1, 3, 5, 10-11, 17-18, 21, 28-31 | Question Bank: 89, 92-93, 113, 145

Question 32
How to handle a differential inequality such as $f'(x) > 2f(x)$?

Model Answer
Bring it to $f' + Pf > 0$ and multiply by IF $= e^{\int P\,dx} > 0$: $\left(f\,e^{\int P\,dx}\right)' > 0$, so $f\,e^{\int P\,dx}$ is increasing; compare with its value at the given point.
Ex: $f' > 2f$, $f(0) = 1$: $(e^{-2x}f)' > 0 \Rightarrow f(x) > e^{2x}$ for $x > 0$.

Grade - Medium
Practice map, main concept: Sheet: J-A 9; C-II 21
Also used in: none

Equations reducible to Linear Differential Equation

Question 33
Write the standard form of Bernoulli's equation.

Model Answer

$$\frac{dy}{dx} + P(x)\,y = Q(x)\,y^n \quad (n \neq 0, 1)$$

Grade - Low
Practice map, main concept: none
Also used in: none

Question 34
What substitution is to be done in Bernoulli's equation?

Model Answer
Divide by y^n and put $z = y^{1-n}$: $\frac{dz}{dx} + (1-n)P\,z = (1-n)Q$ (linear in z).
Ex: $\frac{dy}{dx} = xy + x^3y^2$: $z = \frac{1}{y}$, $\frac{dz}{dx} + xz = -x^3 \Rightarrow z = 2 - x^2 + Ce^{-x^2/2}$.

Grade - Medium
Practice map, main concept: Sheet: J-I 54-55; J-II 41-42; J-M 31; C-II 13 | Question Bank: 146
Also used in: Sheet: J-II 59; J-M 62

Question 35
When to check for change of variable to convert into a linear differential equation?

Model Answer
When the DE has the form $f'(y)\frac{dy}{dx} + P(x)f(y) = Q(x)$: put $f(y) = t$ to get $\frac{dt}{dx} + Pt = Q$ (e.g. $\sec^2 y$ with $\tan y$, e^y with e^y, $\frac{1}{y}$ with $\ln y$).

Ex: $\sec^2 y\frac{dy}{dx} + \tan y = x$: $t = \tan y$, $t' + t = x \Rightarrow \tan y = x - 1 + Ce^{-x}$.

Grade - Medium
Practice map, main concept: Sheet: J-I 57; J-II 45-47; J-M 4, 10, 15; C-I 13, 17; C-II 1, 3, 5 | Question Bank: 93, 113
Also used in: Sheet: J-II 50; J-M 48; C-II 10 | Question Bank: 146

Orthogonal Trajectory

Question 36

Define orthogonal trajectory of a family of curves.

Model Answer

A curve that cuts every curve of the family at a fixed angle α is an isogonal trajectory; if $\alpha = \frac{\pi}{2}$ it is an orthogonal trajectory.

Ex: lines through O are the orthogonal trajectories of circles centred at O.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 37

How to find the orthogonal trajectory of a family of curves?

Model Answer

(i) Form the DE of the family (eliminate the constant). (ii) Replace $\frac{dy}{dx}$ by $-\frac{dx}{dy}$. (iii) Solve the new DE.

Ex: $x^2 + y^2 - 2ay = 0$: $\frac{dy}{dx} = \frac{2xy}{x^2 - y^2}$; trajectories $\frac{dy}{dx} = \frac{y^2 - x^2}{2xy}$, i.e. $x^2 + y^2 = cx$.

Grade - Low

Practice map, main concept: Sheet: C-I 3

Also used in: Sheet: J-II 51

Exact Differential

Question 38

Write three results of exact differential.

Model Answer

$$x\,dy + y\,dx = d(xy), \quad \frac{x\,dy - y\,dx}{x^2} = d\left(\frac{y}{x}\right), \quad \frac{y\,dx - x\,dy}{y^2} = d\left(\frac{x}{y}\right)$$

$$\frac{x\,dy + y\,dx}{xy} = d(\ln xy), \quad \frac{x\,dy - y\,dx}{xy} = d\left(\ln \frac{y}{x}\right), \quad \frac{x\,dy - y\,dx}{x^2 + y^2} = d\left(\tan^{-1} \frac{y}{x}\right)$$

$$\frac{x\,dx + y\,dy}{x^2 + y^2} = d\left(\tfrac{1}{2}\ln(x^2 + y^2)\right), \quad \frac{x\,dy + y\,dx}{x^2y^2} = -d\left(\frac{1}{xy}\right), \quad \frac{ye^x\,dx - e^x\,dy}{y^2} = d\left(\frac{e^x}{y}\right)$$

Grade - Medium

Practice map, main concept: none

Also used in: Sheet: C-II 8 | Question Bank: 139

Question 39

How to solve exact differential based questions?

Model Answer

Group the terms so that each group is a standard exact differential (divide by a suitable factor such as x^2, y^2, xy, $x^2 + y^2$), then integrate term by term. For $yf(xy)\,dx + xg(xy)\,dy = 0$ put $xy = v$.

Ex: $(2xy^2 - y)dx + x\,dy = 0$: $2x\,dx - d\left(\frac{x}{y}\right) = 0 \Rightarrow x^2 - \frac{x}{y} = c$.

Grade - High

Practice map, main concept: Sheet: J-I 56, 60; J-II 43-44, 59; J-M 5, 7, 14, 28-30, 37, 39, 53, 64; J-A 10-11; C-I 10-11; C-II 4, 8-9 | Question Bank: 100, 110, 139

Also used in: Sheet: J-M 34; C-II 31